Experiment 9: TCP Port Scan (Ports 20–1024)

Systems and Network Security Lab

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1 T21 – Nmap

1. Aim

To scan a given host for open TCP ports between 20 and 1024 and display each open port with its commonly associated service name using SuperScan 4.1 (T20), Nmap (T21), and Python sockets (T22).

2. Course Outcome and Module

Course Outcome: CO4.
Syllabus Program 9: Use SuperScan to scan a given host for open TCP ports between 20 and 1024 and display each open port with its service name. T20, T21, and T22 are three versions of this experiment.

3. Tools and Versions

Tool	Version / Environment
Nmap	Version 7.9x as listed in the lab map
Operating system	Kali Linux, Ubuntu, macOS, or Windows
Terminal	System terminal or Command Prompt

4. Program or Commands

Scan TCP ports 20–1024 and skip host discovery:

```
nmap -Pn -p 20-1024 <target>
```

TCP connect scan when not running as root/administrator:

```
nmap -sT -p 20-1024 <target>
```

Check version:

```
nmap --version
```

Replace `<target>` with an authorised hostname or IP address; do not type the angle brackets.

5. How to Run

1. Open a terminal or Command Prompt and run `nmap -version`.
2. Run one of the scan commands against an authorised target.
3. Read the `PORT`, `STATE`, and `SERVICE` columns and capture the output.

The option `-p 20-1024` selects ports 20 through 1024. The option `-Pn` skips host discovery; `-sT` selects a TCP connect scan. Nmap's service names are conventional labels from its service/port database, not proof of the actual software.

6. Sample Output with Screenshots

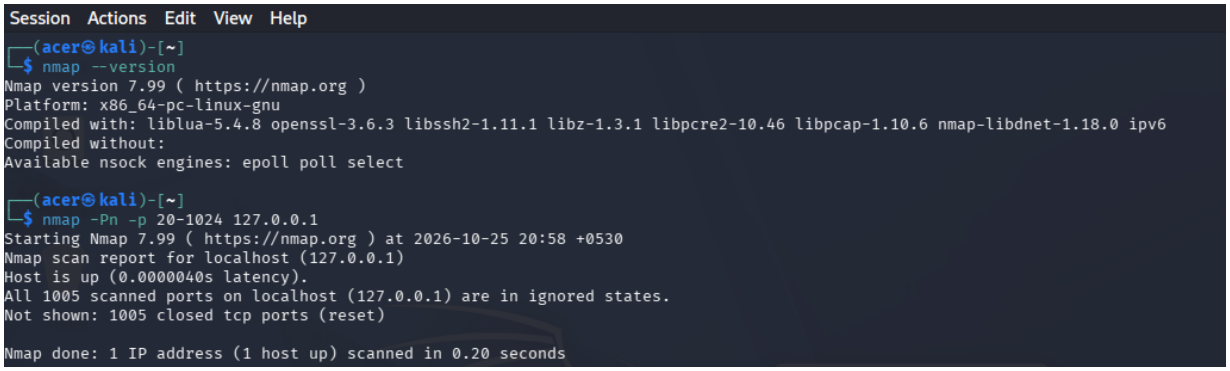


Figure 1: Nmap version verification and basic TCP port scan.

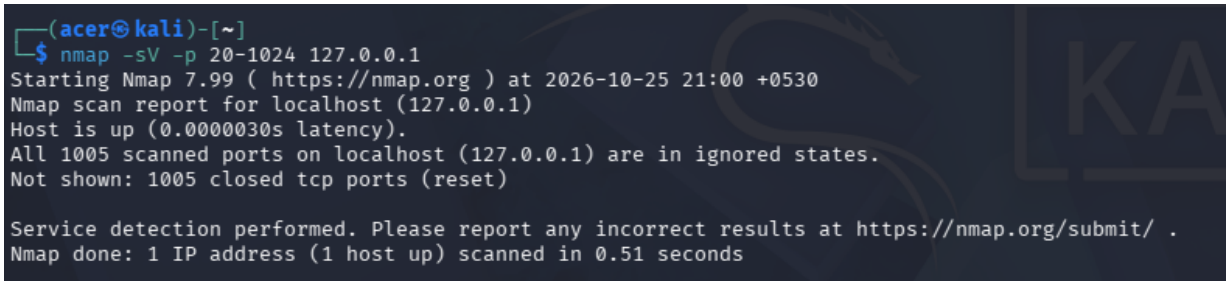


Figure 2: TCP port scan with service detection enabled.

7. Result

Nmap reports the observed state of TCP ports in the selected range and displays conventional service labels.
Target: 127.0.0.1 Open ports observed: 0

8. Difference from Other Versions

Version	Interface	Main distinction
T20 SuperScan	GUI	Tab-based configuration and results
T21 Nmap	Command line	Supports `-Pn`, `-sT`, and port selection
T22 Python	Python program	Scanner logic is implemented using sockets

9. Viva: Five Questions with Answers

1. **What is Nmap?** A tool for network discovery and security auditing.
2. **What does `-p 20-1024` mean?** Scan ports 20 through 1024.
3. **What does `-Pn` do?** It skips host discovery.
4. **What does `-sT` do?** It uses a TCP connect scan.
5. **How are service names displayed?** From Nmap's service/port database.

10. Common Errors: Message, Cause, Fix

Message	Cause	Fix
Host seems down	Discovery probes did not receive a response	For an authorised host, try `-Pn`
Failed to resolve	Incorrect hostname	Check the spelling or use the correct IP
No open ports	No listening service or traffic filtered	Verify target and firewall conditions

Overall Comparison

Task	Method	Service lookup	Distinct feature
T20	SuperScan GUI	Application display	Tab-based graphical workflow
T21	Nmap CLI	Nmap database	Flexible scan flags and port states
T22	Python sockets	`getservbyport()`	Threading and timeout controls

Academic Integrity Note

AI tools were utilized for grammatical review, comparison with alternative versions, assistance in preparing, reviewing, and verifying viva answers, and formatting this report. The results are from the student's practical run. All technical details should be checked against the lab instructions and the course's originality policy.